

After reading introduction, I realized,

the authors identify this main **gap**:

Current nutrition tools are not a good fit for **everyday meal descriptions**. Existing resources are mostly either:

- **tabular food databases**, which require exact food-name matches and can be slow and burdensome for meals with many items, or
- **image-based systems**, which need real-time photos, raise privacy concerns, and can miss hidden food components.

They also say that although natural-language meal descriptions would be much more flexible for users, there was **no existing public dataset/benchmark** to evaluate how well LLMs can estimate nutrition from that kind of input.

Their **solution** is to create **NUTRIBENCH**, which they describe as the **first publicly available benchmark** for nutrition estimation from natural-language meal descriptions. It contains **11,857 human-verified meal descriptions** with labels for carbohydrates, proteins, fats, and calories, built from real dietary intake data across **11 countries**. They then use it to evaluate **12 LLMs** on carbohydrate estimation with different prompting methods like Base, CoT, RAG, and RAG+CoT.

So the gap they found, no public benchmark for testing LLMs on nutrition estimation from normal written meal descriptions. They made a solution and they built NUTRIBENCH and used it to systematically evaluate LLMs on that task.